Architecture Decision Record Process

Document ID	ENG-011
Department	Engineering
Document type	Process
Owner	Principal Architect
Version	4.0
Status	APPROVED
Effective / last reviewed	2025-11-09
Audience	Engineering staff
Related documents	ENG-012, ENG-002, ENG-015

Summary. Explains when Northstar Systems engineers must write an Architecture Decision Record, the template, the Architecture Review Board schedule and voting rules, the four ADR statuses, supersession, and how ADRs link to change, security and open-source reviews.

SYNTHETIC DOCUMENT - Northstar Systems is a fictional company. This document exists for development, testing and demonstration only and is not real legal, employment, security or compliance advice.

1. Purpose and Scope

An Architecture Decision Record (ADR) is a short, permanent document that records one significant technical decision, the context in which it was made, the options considered and the consequences accepted. This process exists so that Northstar Systems engineering can answer, months or years later, the question "why did we build it this way?" without relying on memory, and so that decisions with wide impact are reviewed by people outside the proposing team before they become expensive to reverse.

This process applies to all engineering teams at Northstar Systems Private Limited, in Bengaluru, Mumbai and remote locations, working on the Northstar Nexus Platform, Nexus Connect, Nexus Insight and internal platforms. It covers the decision to write an ADR, the content of an ADR, the review path through the Architecture Review Board (ARB), the status lifecycle and the storage of records. It does not replace the change management process for production deployments (see ENG-002) or the open-source licence review described in ENG-015, although an ADR may trigger or reference either.

The process is intended to be light. A typical ADR is one to two pages and should take an engineer a few hours to write, not days. The purpose is clarity of thinking and a durable record, not bureaucracy.

2. Definitions

- **ADR**: an Architecture Decision Record, written in the standard template and stored in version control.
- **ARB**: the Architecture Review Board, chaired by the Principal Architect, with standing members from Platform Engineering, Site Reliability Engineering, Data Engineering, Security and Product Engineering, plus the VP Engineering as escalation point.
- **Decision driver**: a requirement, constraint or quality attribute that shapes the decision, such as latency, cost, operability or compliance.

- **Reversible decision**: a decision that can be undone in less than one sprint without customer impact.
- **Superseded**: an ADR whose decision has been replaced by a newer ADR.
- **Author**: the engineer who proposes the decision and owns the ADR until it is Accepted or Rejected.
- **Sponsor**: a senior engineer (Band B5 or above) outside the author's immediate team who agrees to champion the ADR in the ARB.

3. Roles and Responsibilities

- **Author**: drafts the ADR, gathers input from affected teams, presents to the ARB and updates the record after the decision.
- **Sponsor**: helps the author prepare, challenges the options and speaks for the ADR in the ARB if the author is a junior engineer.
- **Principal Architect**: owns this process, chairs the ARB, decides agenda order and maintains the ADR index.
- **ARB members**: read submitted ADRs before the meeting, raise concerns in writing where possible and vote.
- **Engineering managers**: ensure their teams raise ADRs when the criteria apply and allow time to implement accepted decisions.
- **Security Champion**: reviews any ADR touching authentication, cryptography or personal data handling.
- **VP Engineering**: resolves deadlocks and approves emergency decisions.

4. When an ADR Is Required

4.1 The two core triggers

An ADR must be written when a decision meets either of these tests.

1 The decision affects more than one team, meaning that another team must change its code, deployment, data or operations, or relies on the outcome.
2 The decision is expected to last more than 12 months, meaning that it would be costly or disruptive to reverse after a year of use.

If a decision meets neither test, an ADR is not required but may be written voluntarily. Teams should record smaller decisions in the pull request description or in the service's design notes.

4.2 Typical examples that require an ADR

- Choosing or replacing a database engine, message broker, cache or search technology for a Tier-1 or Tier-2 service.
- Defining a new public API pattern, authentication model or versioning approach for Nexus Connect.
- Introducing a new programming language or major framework.
- Changing the multi-tenancy model, the data partitioning model or the region topology between Mumbai and Hyderabad.
- Adopting a shared library that several teams will depend on.
- Selecting an external managed service or vendor product that will hold customer production data (the vendor assessment itself follows the third-party security process).
- Deciding to deprecate a public API (the 12-month deprecation notice rule applies in addition).

4.3 Examples that normally do not require an ADR

- Refactoring inside a single service with no interface change.
- Choosing a utility library that is allowed under ENG-015 and used by one team.
- Tuning configuration, such as a thread pool size.
- A reversible experiment limited to a non-production environment.

4.4 Edge cases

When in doubt, the author should ask the Principal Architect in the engineering channel before writing. The rule of thumb is: if you would be annoyed to discover that another team had made this decision without telling you, write an ADR. A decision split into several small pieces to avoid the thresholds above is treated as one decision, and the ARB may require a single ADR to cover it.

A decision that was made informally before this process existed, and that is now being questioned, may be documented retroactively. Such an ADR is marked with the original decision date and the date it was written, and starts in Accepted status if the decision is in force.

5. The ADR Template

Every ADR is a Markdown file named with a sequence number and a short title, for example `0042-adopt-cursor-pagination-for-search.md`, stored in the `architecture/adr` folder of the owning repository or the central architecture repository for decisions that cross repositories. The template has the following sections.

1. **Title**: a short noun phrase describing the decision.
2. **Status**: one of Proposed, Accepted, Superseded or Rejected.
3. **Date and authors**: the date of the last status change and the people involved.
4. **Context**: the problem, the forces at play and the decision drivers. This section should be understandable by an engineer who joins in two years.
5. **Options considered**: at least two realistic options, including "do nothing" where relevant, each with pros, cons and an indicative cost in effort or INR.
6. **Decision**: the chosen option, stated in the active voice ("We will use ...").
7. **Consequences**: the positive and negative results, follow-up work, new risks, effects on operations, and how success will be measured.
8. **Affected teams and services**: who must act.
9. **Review checkpoints**: when the decision should be revisited, for example "after 12 months or 10 million records, whichever is first".
10. **References**: links to design documents, benchmarks, tickets and related ADRs.

Authors should keep the language plain, avoid unexplained abbreviations and attach evidence (benchmark results, cost estimates, proof-of-concept notes) rather than opinion. An ADR that lists only one option will be returned by the ARB.

5.1 Mandatory considerations

The Context or Consequences section must explicitly address each of the following where relevant.

- **Security and privacy**: data classification of the data involved, encryption approach and effect on access control.
- **Operability**: how the solution is monitored (ENG-010), backed up and recovered (ENG-014) and rolled out.

- **Cost**: build cost, running cost per month in INR and licensing cost.
- **Open-source and licensing**: licences of any new dependency against ENG-015.
- **Exit strategy**: how the team would move away from the choice.
- **Technical debt**: any shortcut accepted knowingly. Debt created by an accepted ADR must be recorded so that it can be managed under the technical debt policy (see ENG-012, currently a draft).

6. ADR Statuses and Lifecycle

There are four statuses.

Status	Meaning	Who may set it
Proposed	Written and open for review; the decision is not yet binding	Author
Accepted	Approved by the ARB or by the delegated route in section 7.3; binding on affected teams	Principal Architect after ARB decision
Superseded	Replaced by a newer ADR, which must be linked	Principal Architect, when the new ADR is Accepted
Rejected	Considered and not adopted; retained so the reasoning is not lost	Principal Architect after ARB decision

An ADR is never deleted. A Rejected ADR remains in the repository because a future team may propose the same idea, and the earlier reasoning should save them time. An Accepted ADR may be amended only for typographical corrections or clarifications that do not change the decision. A change of substance requires a new ADR that supersedes the old one.

6.1 Supersession

When a decision is replaced, the new ADR must state which ADR it supersedes and why circumstances changed. The old ADR's status becomes Superseded and gains a link to its successor. Superseded ADRs retain their text so that the history of the architecture can be read in order.

6.2 Expiry of Proposed ADRs

A Proposed ADR that has not been presented to the ARB within 60 days is flagged by the Principal Architect to the author and engineering manager. If it is still not presented within a further 30 days, it is marked Rejected with the reason "withdrawn, no longer needed" unless the author asks to keep it open with a stated reason.

7. Review Path

7.1 Drafting and pre-review

The author writes the ADR in Proposed status and shares the pull request with the affected teams. Affected teams should comment within 5 business days. The author should hold at least one informal discussion with each affected team lead and record the outcome in the Context section. For anything touching security or personal data, the Security Champion must review before the ARB.

7.2 The Architecture Review Board meeting

The ARB meets every second Thursday at 16:00 IST, in person at Northstar Tower, Whitefield, with video conferencing for the Mumbai office at Powai and remote staff. The agenda is published by the Principal Architect on the Monday before the meeting, and ADRs must be submitted by 12:00 IST on the previous Friday of the week before the meeting, giving members at least 4 working days to read them. An ADR submitted late is moved to the next meeting unless the Principal Architect agrees to an urgent slot.

The author presents for up to 10 minutes, followed by up to 15 minutes of discussion. The ARB then decides one of: Accept, Accept with conditions, Request changes (the ADR stays Proposed and returns to a later meeting), or Reject.

A quorum is the Principal Architect or a nominated deputy plus at least 3 other members, of whom at least one must be from outside the author's department head's reporting line. Decisions are by consensus where possible; otherwise by simple majority of members present, with the chair holding a casting vote. A member who has a direct stake in the decision, for example as author's manager, must declare it and abstain.

7.3 Delegated acceptance

For lower-impact ADRs, meaning ADRs that meet only the 12-month test and affect exactly two teams with both team leads in agreement, the Principal Architect may Accept the ADR without a meeting, after at least 5 business days of open comment and no unresolved objection. The decision is reported at the next ARB meeting. Any ARB member may call an ADR into full review within that period.

7.4 Escalation and deadlock

If the ARB cannot reach a decision at two successive meetings, the Principal Architect escalates to the VP Engineering, who decides within 5 business days. If the decision has a significant budget effect, the DoA applies: unbudgeted spend above INR 1,00,000 needs CFO approval in addition to the normal approver, and a purchase above INR 3,00,000 needs a formal RFP through NorthBuy.

7.5 Emergency decisions

In an incident or an urgent security fix, a team may implement a decision before an ADR is reviewed. In that case the team must write the ADR within 5 business days after the emergency, mark it Proposed, and present it at the next ARB meeting. The VP Engineering, or the Engineering Director on call, approves the emergency action. The ARB may require the emergency decision to be reversed.

8. After Acceptance

Once an ADR is Accepted, the author must create tracking tickets in NorthDesk for the follow-up work described in the Consequences section, with owners and due dates. The Principal Architect adds the ADR to the index, which is searchable by service, team, technology and date. Teams must reference the ADR number in the relevant design documents and, where practical, in code comments or configuration notes next to the implementation.

Accepted ADRs are reviewed at the checkpoint stated in the ADR. If no checkpoint is stated, the Principal Architect reviews the ADR 12 months after acceptance for relevance. The review outcome is one of: still valid, to be amended by clarification, or to be superseded. A short note is added to the record.

Teams must not ignore an Accepted ADR. Where a team believes the decision no longer fits, it must propose a superseding ADR instead of silently diverging. The ARB may ask the Principal Architect to include ADR compliance in a service's architecture health check, and findings are recorded as technical debt items.

9. Relationship with Other Processes

- **Change management (ENG-002)**: acceptance of an ADR does not authorise a production change. Every production change still needs a NorthDesk change ticket and a rollback plan, and Normal changes still need Change Advisory Board approval.
- **Open-source review (ENG-015)**: an ADR proposing a library or framework with a licence in the "needs Open Source Review Board approval" category must wait for that approval before being Accepted.
- **Technical debt (ENG-012)**: debt consciously taken on in an ADR must be recorded as a debt item. ENG-012 is currently a draft and is not yet in force, so until it is approved teams should follow their existing backlog practices.
- **Incident learning**: post-incident review action items that require architectural change must be raised as ADRs.
- **Budget**: ADRs that need new spend must name the cost centre, for example CC-ENG-###, and follow the delegation of authority.

10. Worked Examples

Example 1: needs an ADR. The Nexus Connect team wants to move rate limiting from each gateway node to a shared Redis-compatible cache so that the default limit of 1,000 requests per minute per tenant is applied consistently. The change affects Platform Engineering (who run the cache) and SRE (who monitor it) and will be in place for years. It meets both triggers. The author writes ADR 0057, lists three options (per-node counters, shared cache, token service) with monthly running costs of INR 0, INR 85,000 and INR 1,40,000, gets the Security Champion's review for tenant data in the cache, and presents it at the ARB on the second Thursday of the month. The ARB accepts the shared cache with the condition that the failure mode is "fail open with alert". Status becomes Accepted.

Example 2: no ADR needed. An engineer in the Nexus Insight team replaces one CSV parsing library with another MIT-licensed library inside a single reporting service. Only one team is affected and the change is reversible in a day. An ADR is not required; the engineer explains the reason in the pull request and the library choice is checked under ENG-015.

Example 3: supersession. In 2024, ADR 0021 accepted a single-region deployment for Nexus Insight. After the Hyderabad DR region was added, a new ADR 0063 chose active-passive replication. ADR 0063 states that it supersedes ADR 0021, and ADR 0021 changes to Superseded with a link to ADR 0063.

Example 4: rejected ADR retained. A team proposed adopting a document database for Pipeline run history. The ARB rejected the ADR because the operational cost was higher than the benefit and the team lacked on-call experience. The ADR stays in the repository as Rejected. Eight months later a different team proposes the same idea, finds the record, and builds on the earlier benchmarks instead of repeating them.

Example 5: late submission. An author submits an ADR at 15:00 IST on the day of the cut-off, which is after 12:00 IST. The ADR moves to the following ARB meeting, two weeks later, unless the Principal Architect grants an urgent slot because a vendor contract is expiring.

11. Exceptions and Escalation

The Principal Architect may waive the template length or section requirements for simple decisions, but never the Status, Context, Options, Decision and Consequences sections. Decisions that avoid the ARB by being split into small parts are treated as a breach of this process and may be escalated to the VP Engineering. Disagreements about whether an ADR is required are decided by the Principal Architect, with appeal to the VP Engineering.

Questions about the process should go to the Principal Architect through the engineering channel or a NorthDesk request.

12. Frequently Asked Questions

How often does the ARB meet? Every second Thursday at 16:00 IST.

What are the possible ADR statuses? Proposed, Accepted, Superseded and Rejected.

Can I edit an Accepted ADR if I changed my mind? Only for clarifications. A change to the decision requires a new ADR that supersedes the old one.

Is an ADR needed for a decision affecting only my team that will last two years? Yes, because it is expected to last more than 12 months.

What if production is down and I need to decide now? Act under the emergency route, with approval from the Engineering Director on call, and write the ADR within 5 business days.

13. Related Documents

- ENG-002 covers change management and the Change Advisory Board for production changes that implement an accepted ADR.
- ENG-012 is the draft technical debt management policy, not yet in force, which will describe how debt from ADRs is tracked once approved.
- ENG-015 covers open-source and third-party library licences and the Open Source Review Board.
- ENG-010 and ENG-014 describe the monitoring and recovery expectations that ADRs must address.

14. Revision History

Version	Date	Change
3.1	2025-03-06	Added delegated acceptance route
3.2	2025-07-22	Clarified quorum and late submission rules
4.0	2025-11-09	Added emergency decisions, expiry of Proposed ADRs and mandatory considerations